Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/08/2018
PI Name:	Sirmons, Takiyah Ph.D.		
Project Title:	Meal Replacement Mass Reduction and Integration Acceptability Study (Mass Reduction-2)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-244-8443
Organization Name:	KBR/NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/01/2015	**End Date:**	03/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	7
No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmore, Mihriban	**Contact Phone:**	281-244-1004
Contact Email:	mihriban.whitmore-1@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Element change to Human Health Countermeasures; previously Space Human Factors & Habitability (Ed., 1/18/17) NOTE: Change in start date to 1/01/2015 (originally 11/13/2014) per E. Connell/JSC HRP (Ed., 9/16/15)		
Key Personnel Changes/Previous PI:	March 2018: Alexandria Whitmire replaced Jason Schneiderman as the lead investigator from the Behavioral Health & Performance group. December 2016 report: The original statistician, Dr. Robert Ploutz-Snyder, is no longer working on this contract and has been replaced by Dr. Millennia Young. Dr. Thomas Williams has also joined the project and will contribute to analysis of the Human Exploration Research Analog (HERA) survey data. January 2015: The original principal investigator for the study, Monica Leong, is no longer with the contract. Maya Cooper briefly assumed responsibility for the study progress during the period of March 2014 to September 2014. Takiyah Sirmons has since assumed responsibility as principal investigator of this study.		
COI Name (Institution):	Douglas, Grace Ph.D. (NASA) Barrett, Ann Ph.D. (U.S. Army Natick Soldier RD&E Center) Richardson, Michelle B.S. (U.S. Army Natick Soldier RD&E Center) Arias, Diana M.S. (KBRWyle) Schneiderman, Jason Ph.D. (KBRWyle) Young, Millennia Ph.D. (NASA) Slack, Kelley Ph.D. (KBRWyle) Whitmire, Alexandra Ph.D. (KBRWyle)		

Grant/Contract No.:	Directed Research
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NASA has established the goal of traveling beyond low-Earth orbit and extending manned exploration to Mars. The length of proposed Mars missions and the lack of resupply missions increases the importance of nutritional content in the food system, which will need a five year shelf life. The purpose of this research was to assess the stability of vitamin supplementation in traditionally processed spaceflight foods. It was assumed that commercially available fortification nutrients would remain stable through a long duration exploration mission at sufficient levels if compatible formulation, processing, and storage temperatures are achieved. Most bars maintained initial product quality, as evidenced by overall acceptability scores of 6.0 or higher on a 9 point hedonic scale throughout the duration of the study. Likewise, feedback from Human Exploration Research Analog (HERA) crews show that the bars are generally acceptable but additional flavor development is required to combat the effects of food monotony and aversion. The data suggest that meal replacement on an interrupted schedule (every three days) may be acceptable on a mission up to 30 days, but meal replacement every day may result in reduced food intake, greater body mass loss, and increasingly negative changes in mood, affect, and somatic complaints in some individuals. Aims: 1 – Determine methods to increase nutrient density and improve acceptable shelf life of meal replacement prototypes produced in the task “Mass Reduction Technology Development.” 2 - Determine the micronutritional gaps in nutrient dense meal replacement options produced in the task "Mass Reduction Technology Development," and determine fortification strategies. 3 - Determine the microbiological, nutritional, and acceptable sensory shelf life stability of meal replacement options stored at ambient temperature over two years. 4 - Assess and determine an acceptable implementation plan for meal replacement options using analog environments.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal. Currently scheduled Multi-Purpose Crew Vehicle (MPCV) missions are overconstrained by mass, and we are challenged to reduce food system mass by 10% to enable these missions. This is a collaborative study that will efficiently utilize an existing partnership with Natick Combat Feeding Directorate, leveraging their capabilities and expertise, as well as their joint interest in nutrient-dense, true meal replacement development to bring bar development and production knowledge in-house, which will support the flight production that will be required to enable the mass challenges of these missions and future long duration missions. The in-house knowledge will directly and efficiently support HERA studies, to be conducted jointly between AFT and BHP elements, which will ensure that crew continue to be provided with an acceptable food system that supports their health and performance following this mass reduction effort. Given the highly collaborative effort and need for quick turnaround for effective transition of this capability to ops prior to MPCV missions (e.g., EM-2), we propose this as a directed study.
Research Impact/Earth Benefits:	This technology can possibly be applied to military combat feeding or disaster relief efforts.
Task Progress:	The purpose of this task was to develop a variety of nutritionally balanced breakfast replacement bars, which meet spaceflight nutritional, microbiological, sensorial, and shelf-life requirements, while enabling a 10% food mass savings. To date, seven nutrient-dense meal replacement bars containing approximately 700 kcal per serving have been developed. Bars were evaluated in the Human Exploration Research Analog (HERA) to assess the frequency with which actual meal replacement options may be implemented. Feedback from HERA crews show that the bars are generally acceptable but additional flavor development is required to combat the effects of food monotony and aversion. The data suggest that meal replacement on an interrupted schedule (every three days) may be acceptable on a mission up to 30 days, but meal replacement every day may result in reduced food intake, greater body mass loss, and increasingly negative changes in mood, affect, and somatic complaints in some individuals.
Bibliography Type:	Description: (Last Updated: 11/05/2020)
Abstracts for Journals and Proceedings	Whitmire A, Sirmons T, Young M, Schneiderman J, Slack K, Williams T, Douglas G. "Psychological Aspects of Food Bars as a Meal Replacement in Spaceflight." 89th Aerospace Medicine Association Meeting, Dallas, TX, May 6-10, 2018. Aerospace Medicine and Human Performance. 2018 Mar;89(3):218. , Mar-2018
Abstracts for Journals and Proceedings	Whitmire A. "Psychological Aspects of Food Bar/Mass Reduction 2 Meal Replacement Study." Oral presentation at: 2018 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Abstracts for Journals and Proceedings	Sirmons TA. "Meal Replacement Mass Reduction and Integration Acceptability Study." Oral presentation at: 2018 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
Abstracts for Journals and Proceedings	Sirmons TA. "Meal Replacement Mass Reduction and Integration Acceptability Study." Oral presentation at: 2017 Annual Meeting for the Institute of Food Technologists, Las Vegas, NV, June 26-28, 2017. 2017 Annual Meeting for the Institute of Food Technologists, Las Vegas, NV, June 26-28, 2017. , Jun-2017
Articles in Peer-reviewed Journals	Sirmons TA, Roma PG, Whitmire AM, Smith SM, Zwart SR, Young M, Douglas GL. "Meal replacement in isolated and confined mission environments: Consumption, acceptability, and implications for physical and behavioral health." Physiol Behav. 2020 May 15;219:112829. https://doi.org/10.1016/j.physbeh.2020.112829 ; PMID: 32068108 , May-2020